The empirical recurrence for OEIS A267906, proved by transfer matrix

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Abstract

OEIS A267906 counts arrays in which the REPEATED VALUES of each row, and of each column, stand in a prescribed relation to one another in the order they occur. The entry carries an empirical recurrence of order 7 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: one of the two scans runs inside a slice of the array and the other across slices, and the second needs only the previous slice together with the last repeated value so far in each column of the scan, so the count is a walk count on $S = 185$ states.

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1 The conjecture

OEIS A267906 is "Number of n X 2 0..2 arrays with every repeated value in every row unequal to the previous repeated value, and in every column equal to the previous repeated value, and new values introduced in row-major sequential order.". It has offset 1 and begins

$$2, 14, 122, 938, 6734, 45938, 302402, 1939154, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 14*a(n-1) - 60*a(n-2) + 50*a(n-3) + 145*a(n-4) - 80*a(n-5) - 84*a(n-6) + 16*a(n-7).

As of the "Last modified" line on the live entry (Feb 25 2018 08:39:08, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Read a row of the array from left to right. A REPEATED VALUE is an entry equal to the entry immediately before it; the row thus produces the sequence of values at the places where two equal entries stand side by side. The entry asks that each repeated value of a row be different from the previous repeated value of that row, and that each repeated value of a column, read downwards, be equal to the previous repeated value of that column. The FIRST repeated value of a scan has no predecessor and is unconstrained; a scan with no two equal neighbours imposes nothing at all.

The entry closes with "new values introduced in row-major sequential order", so what is counted is arrays up to renaming the letters. Both relations here are built from equality alone, so the condition survives renaming and the usual inversion applies: with N_j the classes on exactly j letters and L_i the admissible arrays over an alphabet of i letters,

$$L_i = \sum_j N_j \, i(i-1)\cdots(i-j+1), \qquad a = N_1 + \cdots + N_3 = \big(\mathbf{1}^\top F^{-1}\big)L.$$

An entry of this family whose relation compares letters by size, or does arithmetic on them, carries no such clause and is counted over its own alphabet instead; the two cases are not mixed.

3 One scan inside a slice, one across

Take the array a *row* at a time, the direction in which it grows.

Lemma 1. *The row scans are settled as each row is written. The column scans are settled by the previous row together with, for each position, the last repeated value so far in that column.*

Proof. A *row* scan compares neighbours inside one *row*, so writing the *row* left to right and carrying the last repeated value seen decides it. In a *column* scan a repeat happens at the new *row* exactly when the new entry equals the entry in the same position of the previous *row*, and the only thing the relation needs besides that entry is the previous repeated value of that *column*. □

So the state is the pair (previous *row*, vector of last repeated values), the second component taking $2+2$ values at each position — the letters together with a symbol for "none yet". Assembling the pieces for $i = 1, \ldots, 3$ as a disjoint union, with ι carrying the weights above cleared of denominators,

$$a(n) \;=\; \iota^\top M^{\,j}\tau, \qquad S = 185, \quad \tau = \mathbf{1}.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^7 - \sum_i c_i t^{7-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+7} - \sum_i c_i M^{j+7-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 14\,a(n-1) - 60\,a(n-2) + 50\,a(n-3) + 145\,a(n-4) - 80\,a(n-5) - 84\,a(n-6) + 16\,a(n-7)$$

for every $n > 7$.

Proof. The vector $w = q(M)\tau$ was formed by 7 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 7+1$ onwards and the run continued until the working vector was identically zero. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, form each row's and each column's list of repeated values by scanning it, and test consecutive members of those lists against the relation. No state, no window and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A267906.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.